Quantum Variational Algorithms on Adabatic Quantum Computing Devices

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1 Overview

Since the development of the quantum computers, many claims regarding quantum advantage have been made. Often, these advantage claims did not last long. In this work, we focus exclusively on the D-Wave Quantum Annealer, which leverages quantum technology to solve quadratically binary unconstrained optimization (QUBO) problems. D-Wave is among the first companies to commercially sell quantum computers. In 2025, researcher from D-Wave claimed to have solved a scientifically relevant problem fast using quantum technology, than it would have been possible on classical devices. However, it is currently an open question whether the D-Wave Quantum Annealer is able to solve other problems well, and possible faster than convential computers. In this work, we assess the D-Wave's ability to sample from a probability distribution, to approximate the ground state of different statistical mechanical models using variational algorithms.

1.1 Previous work & motivation

We give an overview of the current research on quantum computing devices and present our motivation for using variational algorithms to simulate quantum systems.

1.1.1 Quantum Annealing Performance

The D-Wave's Quantum Annealers performance scaling for finding the ground state (corresponding to the minimum of the quadratic binary cost function) has been extensively studied, in the area of cryptography and physical simulation. In general, classical methods exhibit faster solving times and better scaling. We briefly report on previous work in the field of simulating physical systems using quantum annealing hardware. In [1], we used a method presented in [2] to encode the solution to the wave function of different Hamiltonians into a QUBO problem. These instances were then solved on different physical und physics-inspired solvers. We evaluate the scaling of the samplers using the time to solution (TTS) metric, which corresponds to the computational time needed to solve the optimisation problem with a success probability of at least P_{target}. The TTS is defined as

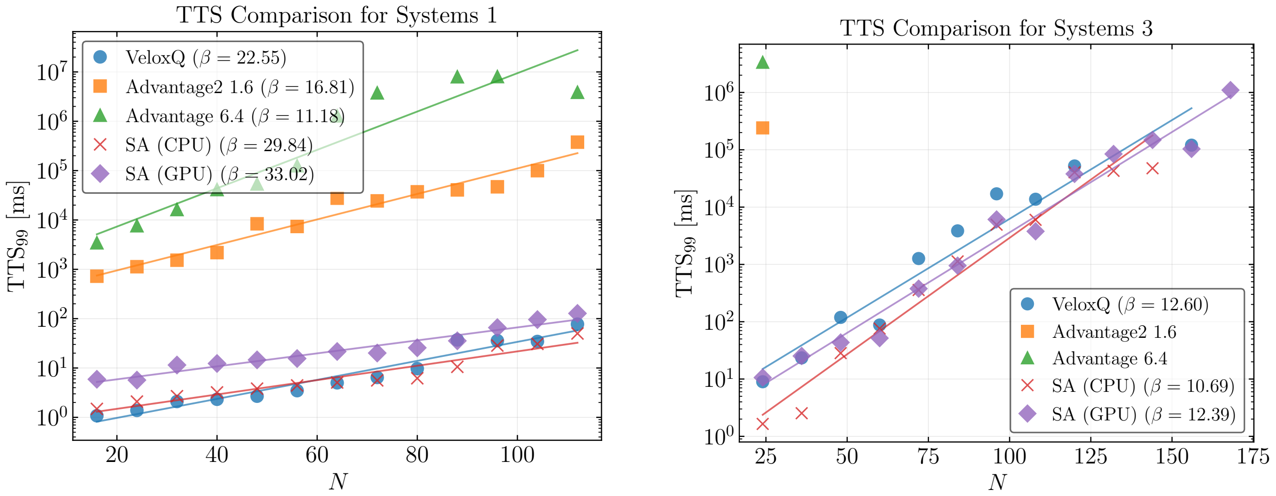
$$\text{TTS} = \frac{\ln(1 - P_{\text{target}})}{\ln(1 - P_{\text{success}})} T_r, \tag{1}$$

where P_{success} is the probability of obtaining the ground state in a single run, and T_r denotes the runtime of a single run.

It is important to note that current D-Wave devices exhibit limiting connectivity. As a consequence, most problem instances need to be adapted in an additional *embedding* step, increasing the number of physical qubits required. We compare two problem formulations, one advantageous for the quantum annealer, as no extra embedding step is required, and a second one after embedding. Classical devices dominate in both cases (see Figure 1).

1.1.2 Motivation

As previously shown, results return by the quantum annealer are inherently noisy. This makes it difficult to sample the *exact* ground state of the submitted Hamiltonian. In this work, we therefore assess the device's ability to act as a Gibbs sampler and sample from the Boltzmann distribution. It has been shown earlier that the D-Wave device can in fact be used as a Gibbs sampler[3][5]. Here, we systematically evaluate the quality and time of these samples on computationally hard problems, which allows us to rigorously compare the quantum annealer to classical physics inspired algorithms.



(a) D-Wave native formulation. This corresponds to a single-qubit Y rotation, governed by $H_1 = \frac{\pi}{2}\sigma_y$.

(b) Non-native formulation. This corresponds a single-qubit Rabi oscillations driven along y, described by $H_3 = \pi\left(\frac{3}{4}\mathbb{1} - \frac{1}{4}\sigma_y\right)$.

Figure 1: Previous work on sampling the exact ground state to simulate quantum systems. TTS99 vs. N: exponential scaling $\propto \exp(N/\beta)$. GPU solvers dominate for $N \gtrsim 100$; missing points indicate zero success probability.

2 Methods

In this section we briefly outline the simulated physical systems as well as the main computational and algorithmic methods used in this work.

2.1 Quantum Ising Models

The Ising model consists of discrete spins arranged on a lattice, interacting via nearest-neighbour couplings. In its quantum extension, the Transverse Field Ising Model (TFIM), a transverse magnetic field is added, introducing quantum fluctuations through non-commuting terms in the Hamiltonian.

2.1.1 Transverse Field Ising Model

The Hamiltonian for the one-dimensional TFIM is given by

$$H = -J \sum_i \sigma_i^z \sigma_{i+1}^z - h \sum_i \sigma_i^z \tag{2}$$

where J is the coupling strength and h is the transverse magnetic field strength. The 1D TFIM serves as a benchmark for our models, as it is exactly solvable via the Jordan–Wigner transformation. Despite its simplicity, it displays non-trivial quantum behavior, including a quantum phase transition and quantum fluctuations.

2.1.2 Long-range Transverse Field Ising Model

The Hamiltonian for the one-dimensional long-range transverse-field Ising model (LRTFIM) is given by

$$H = -\sum_{i<j} \frac{J}{|i-j|^\alpha} \sigma_i^z \sigma_j^z - h \sum_i \sigma_i^x \tag{3}$$

where J sets the interaction strength, α controls the power-law decay of the interactions, and h is the transverse magnetic field strength. The model serves as an extension of the 1D-TFIM and allows us to assess the ability of our models to capture long-range interactions.

2.2 Heisenberg Models

The Heisenberg model describes spins on lattice sites interacting via exchange coupling:

$$H = \sum_{\langle i,j \rangle} J_{ij}\, \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j \tag{4}$$

where $\boldsymbol{\sigma}_i = (\sigma_i^x, \sigma_i^y, \sigma_i^z)$ are the Pauli spin operators, J_{ij} is the exchange coupling between sites i and j, and $\langle i, j \rangle$ denotes nearest-neighbour pairs. In contrast to the TFIM, the Heisenberg model couples all three spin components.

2.2.1 XXZ Heisenberg Chain

We study the one-dimensional XXZ Heisenberg chain with periodic boundary conditions:

$$H = J \sum_{i=1}^{N} \left[\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y + \Delta\, \sigma_i^z \sigma_{i+1}^z \right], \tag{5}$$

where J is the exchange coupling and Δ is the anisotropy parameter. Setting $\Delta = 1$ gives the isotropic Heisenberg model, while $\Delta = 0$ reduces to the XY chain. Unlike the TFIM, the model couples all three spin components, making it a more general and challenging benchmark.

2.2.2 Frustrated J_1-J_2 Heisenberg Chain

The frustrated J_1-J_2 Heisenberg chain extends the XXZ model with a next-nearest-neighbour exchange term:

$$H = J_1 \sum_i \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+1} + J_2 \sum_i \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+2}, \tag{6}$$

where J_1 and J_2 are the nearest- and next-nearest-neighbour coupling strengths. The competing interactions frustrate the system, preventing simultaneous minimisation of both exchange terms. Setting $J_2 = 0$ recovers the standard Heisenberg chain, while increasing J_2/J_1 drives the system into a more frustrated regime that poses a more stringent test for variational methods.

2.3 Quantum (inspired) solvers

Here we present two solver types: First, a special quantum machine designed to find the ground state of an Ising Hamiltonian, and second, a classical quantum inspired algorithm.

2.3.1 Adiabatic Quantum Computing

Adiabatic Quantum Computing is based on the adiabatic theorem, which states: *A physical system remains in its instantaneous eigenstate if a given perturbation is acting on it slowly enough and if there is a gap between the eigenvalue and the rest of the Hamiltonian's spectrum.* In this work we focus on the D-Wave Quantum Annealer, which acts according to the adiabatic theorem in the following way: In the beginning of the annealing process, the device is initiated in the ground state of a initial driver Hamiltonian H_D that can be easily constructed:

$$H_D = -\sum_{i=1}^{N} \sigma_x^{(i)} \text{ with ground state } |+\rangle^{\otimes N} \tag{7}$$

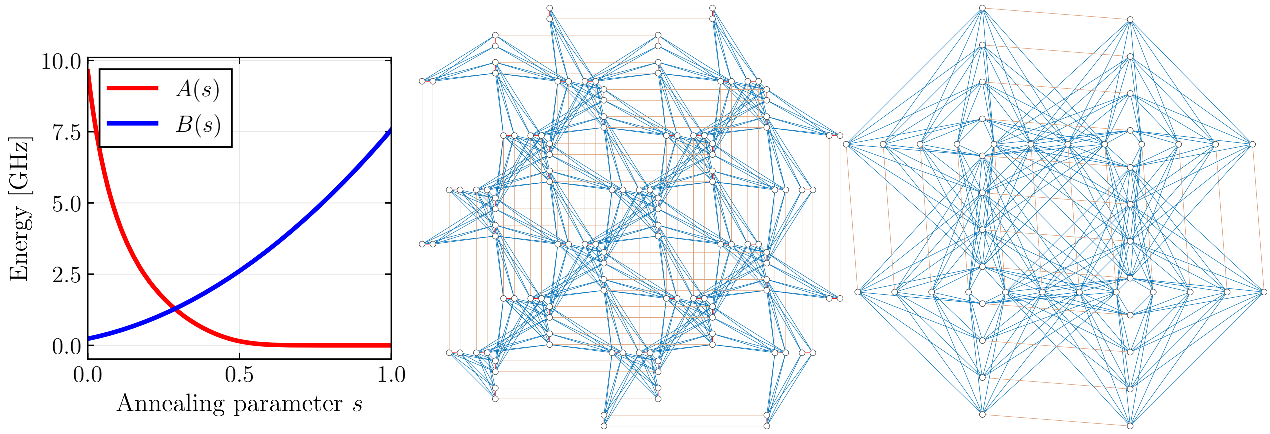
where σ_x, σ_y, and σ_z are the Pauli matrices acting on qubit i. The problem Hamiltonian is defined as

$$H_P = \sum_{i=1}^{N} = h_i \sigma_z^{(i)} + \sum_{i,j=1}^{N} J_{ij} \sigma_z^i \otimes \sigma_z^j. \tag{8}$$

During the annealing process, the system is slowly brought from H_D to H_P according to the so-called annealing schedules $A(s)$ and $B(s)$:

$$H(s) = A(s)H_D + B(s)H_P = -A(s) \sum_{i=1}^{N} \sigma_x^{(i)} + B(s) \sum_{i=1}^{N} h_i \sigma_z^{(i)} + B(s) \sum_{i=1}^{N} \sum_{j=1}^{i-1} J_{ij}\, \sigma_z^{(i)} \otimes \sigma_z^{(j)}, \tag{9}$$

where $s \in [0, 1]$ is the normalized time. $A(s)$ decreases the driving terms in H_D as s increases, and $B(s)$ activates the terms of the problem Hamiltonian H_P. At the end of the evolution, if the process happens slow enough, and assuming a noiseless environment, the system will be in the ground state of the problem Hamiltonian according to the adiabatic theorem.



(a) (b) (c)

Figure 2: (a) Time-dependent energy scales executed by the D-Wave quantum annealer. The coefficient $A(s)$ (red) multiplies the transverse-field term that enables quantum tunnelling and therefore falls rapidly as the run progresses. The coefficient $B(s)$ (blue) multiplies the problem Hamiltonian and rises smoothly, so that the device moves from a tunnelling-dominated regime at $s = 0$ to a problem-dominated regime at $s = 1$. (b) Native connectivity map for the Pegasus architecture (D-Wave Advantage). Circles mark individual superconducting qubits; blue lines are internal couplers and orange lines mark external couplers (c) Connectivity map for the newer Zephyr architecture (D-Wave Advantage2), which increases the number of couplers per qubit and introduces longer-range links.

2.3.2 Simulated Bifurcation

Simulated Bifurcation (SB) is a heuristic algorithm that numerically simulates adiabatic and ergodic (chaotic) evolutions of classical nonlinear Hamiltonian systems to solve combinatorial optimization problems. Inspired by quantum adiabatic optimization using networks of Kerr-nonlinear parametric oscillators (KPO), the algorithm encodes the ground state of an Ising Hamiltonian into the stable states of a network of classical oscillators.

Each Ising spin is represented by continuous dynamical variables, position x_i and momentum y_i, which serve as canonical conjugate variables for each oscillator. The system evolves according to coupled differential equations derived from a network of Duffing oscillators. These equations include a nonlinear x^4 term and a time-dependent pumping parameter $p(t)$. As $p(t)$ is gradually increased, the system undergoes an adiabatic bifurcation, in which a single stable equilibrium at the origin splits into two stable branches. This mechanism drives the trajectories toward configurations corresponding to low-energy states of the Ising Hamiltonian. Because all variables are updated simultaneously at each time step, the algorithm is highly parallelizable and well suited for implementation on FPGAs and GPUs.

Langevin Simulated Bifurcation Langevin Simulated Bifurcation (LSB) is a stochastic adaptation of SB designed to function as a Boltzmann sampler for training energy-based models such as Semi-Restricted Boltzmann Machines (SRBMs). While SB is a deterministic optimization method, LSB introduces stochasticity to approximate a Boltzmann distribution $B_{\beta_{\text{eff}}}$ with an unknown effective inverse temperature β_{eff}. To handle the unknown β_{eff} during learning, LSB is often combined with Conditional Expectation Matching (CEM). A detailed analysis of different temperature parameters is provided in subsection 4.3.

Conditional Expectation Matching CEM accounts for the fact that, in order to produce samples using LSB, a parameter β_{eff}, which is unknown and instance dependent, has to be found. This parameter is called the effective inverse temperature and it defines the distribution returned by the sampler. In Semi-Restricted Boltzmann Machine (SRBM), hidden variables are conditionally independent when the visible units are fixed to a condition vector r. This allows the direct calculation of the expectation value of a hidden unit:

$$\langle h \rangle_{\beta,r} = \tanh\left\{\beta\left(c_j + \sum_{i=1}^{N_v} r_i W_{ij}\right)\right\} \tag{10}$$

To estimate the expectation value, denoted $\langle h_j \rangle_{r,C}$ CEM then performs conditional LSB sampling, fixing $v = r$. In the *matching* step, CEM then minimizes the squared difference between both expectation values:

$$\beta_{\text{eff}} = \arg\min_{\beta>0} \sum_{j=1}^{N_h} \left\{\langle h_j \rangle_{r,C} - \langle h \rangle_{\beta,r}\right\}^2 \tag{11}$$

CEM can also be used with other temperature-dependent solvers, e.g. the D-Wave Quantum Annealer.

2.4 Metropolis-Hastings inspired methods

We use several Markov chain Monte Carlo (MCMC) methods to generate samples from probability distribution from which direct sampling would be difficult. All of these methods can be traced back to the Metropolis-Hastings algorithm, which we explain below.

2.4.1 Metropolis-Hastings Algorithm

The Metropolis-Hastings Algorithm is a MCMC method used to obtain samples from a (multivariate) probability distribution $P(x)$ from which sampling is difficult. The goal of the algorithm is to define a Markov chain whose stationary distribution is $P(x)$. Suppose we can evaluate an unnormalized target density $f(x)$, where $f(x) \propto P(x)$, and the normalization constant is unknown. In the **proposal** step, a *candidate* x^* from a proposal distribution $q(x^*|x_n)$ is generated, x_n being the current state of the Markov chain. In the **accept-reject** step, the acceptance probability $A(x_n \to x^*)$ is calculated[4]:

$$A(x_n \to x^*) = \min\left(1, \frac{f(x^*)}{f(x^n)} \times \frac{q(x_n|x*)}{q(x^*|x_n)}\right). \tag{12}$$

For the evaluation of $\frac{f(x^*)}{f(x^n)}$, knowledge of the normalization constant is not necessary. The candidate x^* is accepted with probability $A(x_n \to x^*)$.

2.4.2 Simulated Annealing

simulated annealing (SA) is the classical analogue and predecessor of quantum annealing. It works by exploring the neighborhood of a given configuration, accepting the proposed solution with a temperature dependent probability, or if the new configuration has a lower energy than the current one.

2.4.3 Gibbs sampling

Gibbs sampling can be considered a special case of the Metropolis-Hastings algorithm. It is used when sampling directly from a joint distribution is difficult, but sampling from the full conditional distributions is tractable. The algorithm constructs a Markov chain by iteratively sampling each variable (or block of variables) from its distribution conditioned on the current values of all other variables. This procedure generates a chain whose stationary distribution is the target joint distribution.

3 Variational Algorithms

Variational algorithms are a class of hybrid quantum-classical optimization methods. We focus on the Variational Quantum Eigensolver (VQE) which is designed to find the ground state of a given Hamiltonian. It consists of three parts: the *cost function* to be minimized, the *ansatz*, which in our case is a Boltzman machine, and an *expectation value estimation*. Additionally, a classical optimizer is needed to improve the ansatz at each iteration. Depending on the quantum architecture, two approaches for estimating the expectation value are possible:

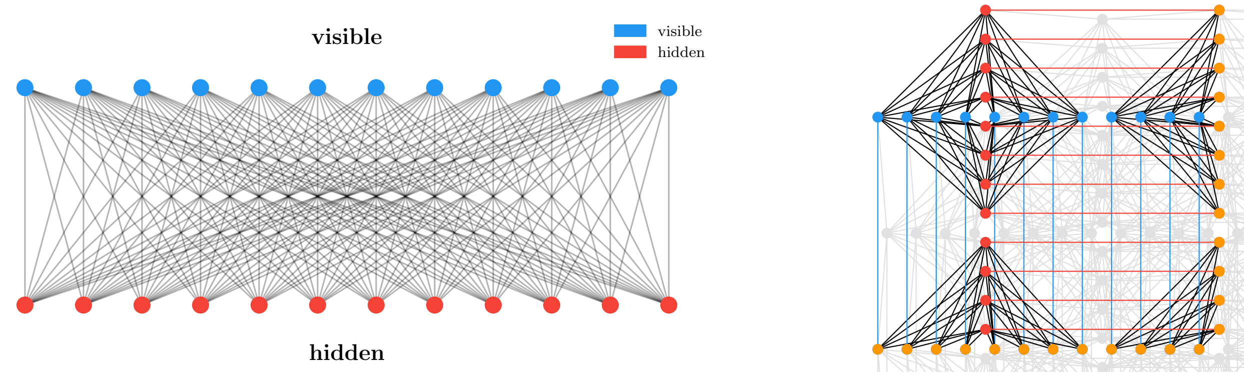
- On gate-based quantum computers, measuring can be used to gather expectation values of the modified ansatz and improve it further
- In adiabatic quantum computing, the quantum device is used to generate representative samples of the modified ansatz. These samples are then used to approximate the expectation value.

As we are interested in comparing current quantum annealing devices with their classical counterparts, we focus solely on the second case. The goal is to approximate the ground state, that is, the eigenstate corresponding to the lowest eigenvalue of a Hamiltonian H. In the adiabatic case, the ansatz can be written as

$$|\Phi\rangle = \sum_v \Phi(v) |v\rangle \text{ , with } \Phi(v) = \sum e^{-\Phi(v,h)} \tag{13}$$

with $\Phi(v,h) = \boldsymbol{a} \cdot \boldsymbol{v} + \boldsymbol{b} \cdot \boldsymbol{h} + \boldsymbol{h} \cdot \boldsymbol{W} \cdot \boldsymbol{v}$. The network is specified by the weights $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{W}$, which are respectively applied to the vector of visible units $\boldsymbol{v} = (v_1, \ldots, v_N)$ and hidden units $\boldsymbol{h} = (h_1, \ldots, h_N)$. The ansatz is defined as

$$\Psi(v) = \sqrt{\sum e^{-\Phi(v,h)}} = \sqrt{\Phi(v)} \tag{14}$$



(a)

(b)

Figure 3: (a) RBM with 24 visible and 24 hidden units. (b) Embedding of the corresponding bipartite graph onto the Zephyr hardware topology. Blue nodes are visible units, red nodes are hidden units, orange nodes are auxiliary chain qubits. Each physical node is connected to one auxillary qubit.

with probability distribution

$$\rho(v) = \frac{|\Psi(v)|^2}{\sum_{v'} |\Psi(v')|^2} \tag{15}$$

which is however intractable for large Hilbert spaces.

3.1 Boltzman machine

A restricted Boltzmann machine (RBM) is an artificial neural network consisting of visible and hidden units that form a bipartite graph. That is, no inter-layer connections are allowed. During training, the RBM learns to approximate a probability distribution. The main idea is that expectation values with respect to the quantum state can be estimated using samples drawn from the probability distribution $|\Psi(v)|^2$. Since the wavefunction ansatz is represented by a Restricted Boltzmann Machine, which corresponds to a classical stochastic Ising model, configurations v can be sampled approximately using a D-Wave quantum annealer.

4 Experimental Analysis

We conduct experiments analyzing sampler quality and physical properties or samplers and models.

4.1 Sampling Quality

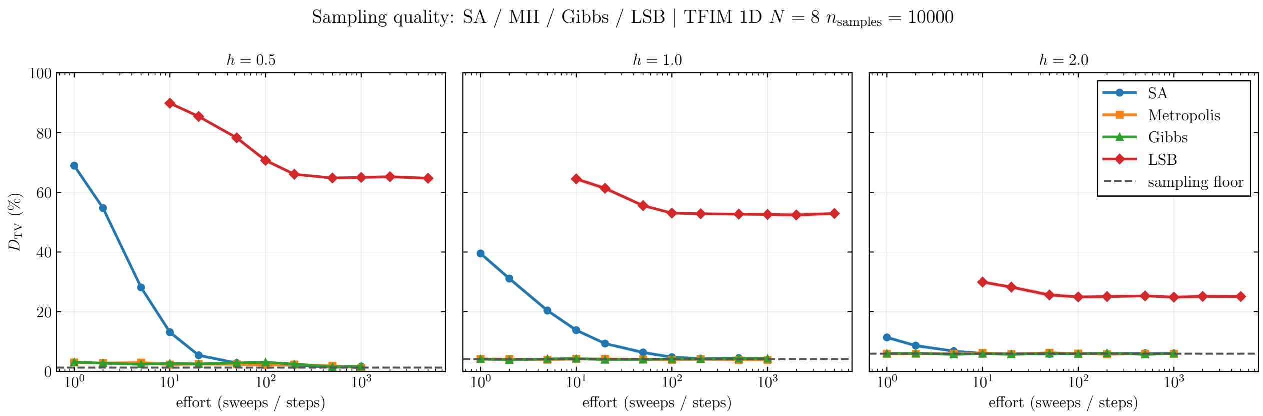
While some of the samplers are guaranteed to sample from the Boltzman distribution, others are not. In this section, we analyze the quality of samplers with regard to various parameters. One measure for sampling quality is the *total variation distance* D_{TV}. It is defined as

$$D_{\text{TV}}(\mu, \nu) = \frac{1}{2} \sum_{\boldsymbol{\sigma} \in \{-1,+1\}^N} |\mu(\boldsymbol{\sigma}) - \nu(\boldsymbol{\sigma})| \,, \tag{16}$$

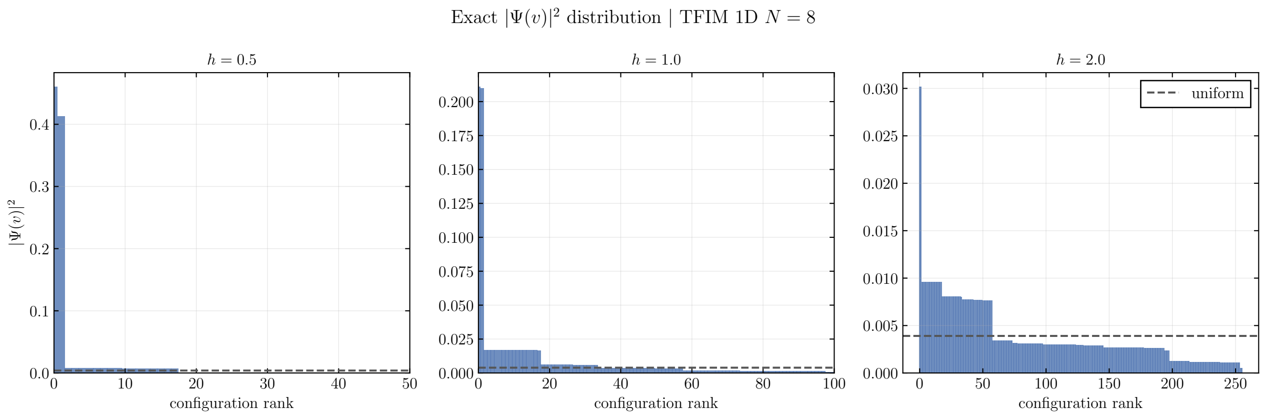
where μ is the histogram returned from the sampler and $\nu(\boldsymbol{\sigma}) \propto |\Psi(\boldsymbol{\sigma})|^2$ is the RBM distribution. As every sampler has a range of parameters, we are interested in how well the sampling performs with regards to the different tuning parameters. SA, Gibbs, and Metropolis can be tuned by the number of *sweeps*, that is, the number of single-spin-flip proposals, whereas LSB is tunable by the number of *steps* which updates all spins simultaneously.

We use the following workflow to determine D_{TV}: First, we use a SA pretrained RBM and compute its exact distribution $|\Psi|^2$ numerically by enumerating all possible spin configurations. We then collect samples from different samplers and compare the resulting empirical histogram to the exact distribution.

The results are presented in Figure 2a. As a sanity check, it can be seen that Metropolis and Gibbs sampling perform theoretically near-optimal, as is expected. We observe that LSB significantly underperforms the other solvers for this small-scale example. The performance of both LSB and SA improves as the transverse field bias h increases. In Figure 4**(b)** the exact histogram of configurations $\Psi(v)|^2$ is plotted. For $h = 0.5$ the distribution



(a) Total variation distance as a function of sampling effort.



(b) Exact histogram of configurations.

Figure 4: Sampling quality for different samplers for a trained RBM approximating the ground state of a TFIM with 8 spins, corresponding to a RBM with 8 visible and 8 hidden units. The sampling floor is the best possible total variational distance reachable with the given number of samples. Effort is measured in sweeps for SA, metropolis and Gibbs, and in steps for LSB.

is peaked around a handful of highly probable sample configuration, whereas the distribution becomes more uniform for stronger values of h. This explains the performance increase for LSB and SA samplers.

The peaked distribution originates from the fact that the spins in the TFIM description given by Equation 2 are aligned for low h values (*ferromagnetic* phase). For $h \to \infty$ the transverse field dominates and the ground state becomes a product of single-spin σ_x eigenstates. Measured in the z-basis, this leads to a uniform distribution in the thermodynamic limit.

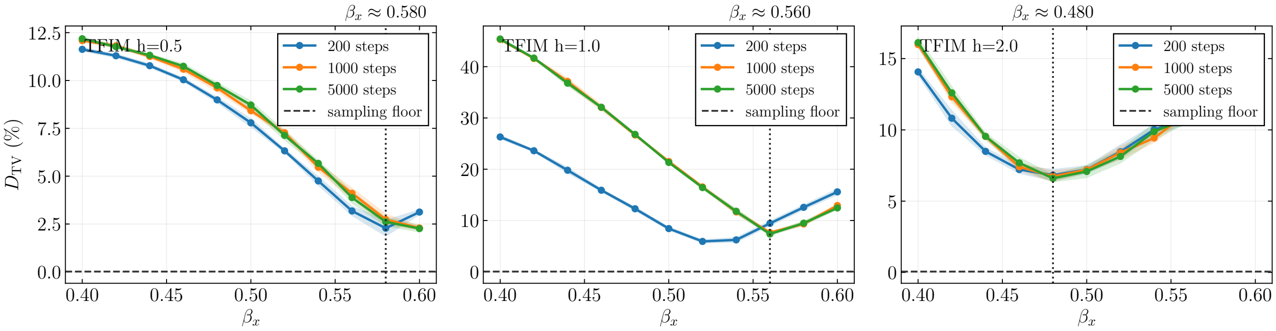
4.2 Sampling temperature

In this section, we evaluate the influence of temperature on different models and solvers. We focus on two quantities: the rescaling factor β_x and the effective temperature β_{eff}. Sampling from an ansatz, all weights in the RBM are rescaled by β_x. For an ideal Gibbs sampler, the following equation holds:

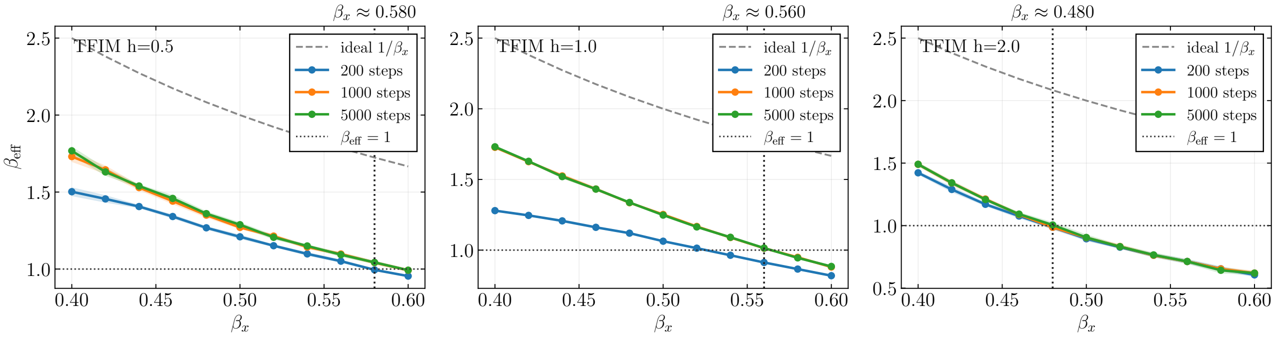
$$\beta_{\text{eff}} = \frac{1}{\beta_x}. \tag{17}$$

4.3 Langevin Simulated Bifurcation

As LSB is a temperature dependent solver and not guaranteed to sample from the Boltzmann distribution, it can be used to illustrate the influence of the input temperature β_x on the solution quality. Contrary to the quantum annealer, sampling using LSB does not involve additional costs. To evaluate the inpact of the scaling parameter β_x, we calculate the total variational distance D_{TV} between the samples and the precise Boltzmann distribution by numerical enumeration. Figure 5 clearly illustrates that the sampler temperature is problem dependent. D_{TV} is minimal at the point where the effective temperature β_{eff} is 1.



(a) Total variation distance D_{TV} between the LSB empirical histogram and the exact distribution $|\Psi(v)|^2$. The horizontal dashed line marks the sampling floor.



(b) Effective inverse temperature β_{eff} of the LSB samples, estimated by minimizing the total variational distance between the samples and the actual distribution. The dashed curve shows the ideal $1/\beta_x$ prediction for a perfect Gibbs sampler, the horizontal dotted line marks the VMC target $\beta_{\mathrm{eff}} = 1$.

Figure 5: Influence of the LSB energy-scale parameter β_x on sampling quality for a trained TFIM RBM with $N = 8$ visible spins. The vertical dotted line in each panel marks the β_x that minimises D_{TV}.

4.3.1 CEM

4.4 Limitations of real-valued RBMs

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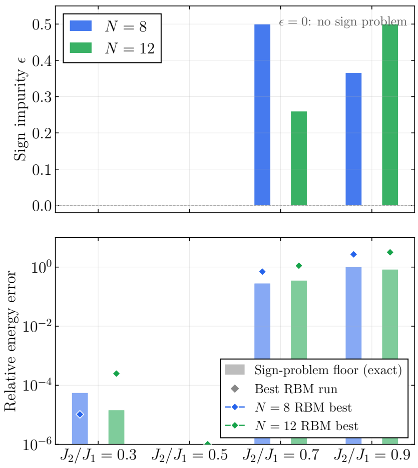
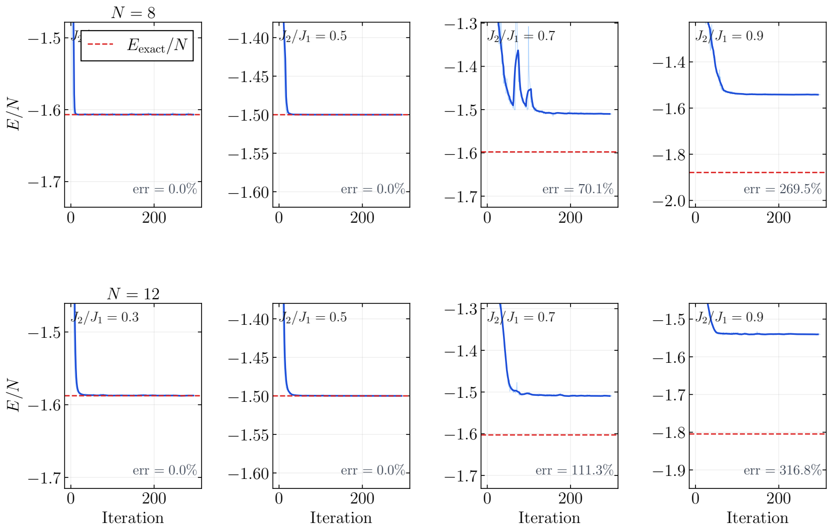


(a) Sign impurity ϵ (top) and theoretical energy floor for any real positive ansatz vs. best RBM error (bottom), computed by exact enumeration at four values of J_2/J_1 for $N = 8$ and $N = 12$. The floor equals zero for $J_2/J_1 \leq 0.5$ (Marshall theorem).



(b) Best RBM convergence curves (E/N vs. SR iteration) for $N \in \{8, 12\}$ and $J_2/J_1 \in \{0.3, 0.5, 0.7, 0.9\}$. The red dashed line marks the exact ground-state energy E_{exact}/N. Before and at the Majumdar–Ghosh point ($J_2/J_1 \leq 0.5$) the RBM converges to the ground state (relative error 0.0%). Beyond it the optimiser stalls well above the exact energy, with the gap growing with J_2/J_1 and with system size N.

Figure 6: The sign problem as the fundamental barrier for real-valued RBMs on the frustrated J_1–J_2 Heisenberg chain. Left: exact-enumeration proof that the barrier is a representational limitation, not an optimisation failure. Right: the corresponding training dynamics.

References

[1] Philipp Hanussek, Jakub Pawłowski, Zakaria Mzaouali, and Bartłomiej Gardas. Quantum-inspired dynamical models on quantum and classical annealers. *Phys. Rev. Appl.*, 25:044055, Apr 2026.

[2] Konrad Jałowiecki, Andrzej Więckowski, Piotr Gawron, and Bartłomiej Gardas. Parallel in time dynamics with quantum annealers. *Scientific Reports*, 10(1):13534, 2020.

[3] Vrinda Mehta, Hans De Raedt, Kristel Michielsen, and Fengping Jin. Understanding the physics of d-wave annealers: From schrödinger to lindblad to markovian dynamics. *Physical Review A*, 112(3), 2025.

[4] Danielle Navarro. The Metropolis-Hastings algorithm. `https://blog.djnavarro.net/posts/2023-04-12_metropolis-hastings/#fnref1`. [Accessed 16-06-2026].

[5] Jon Nelson, Marc Vuffray, Andrey Y. Lokhov, Tameem Albash, and Carleton Coffrin. High-quality thermal gibbs sampling with quantum annealing hardware. *Physical Review Applied*, 17(4), April 2022.